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ADBMS-Expt 6

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Question 1

Experiment 06

Medium

(A) Problem Statement

A fruit distribution company records the number of apples and oranges sold each day. For every date, exactly one record exists for apples and one record exists for oranges.

Generate a daily sales comparison report showing the difference between the number of apples sold and the number of oranges sold on the same day. The difference should be calculated as:

Difference = Apples Sold - Oranges Sold

Display the sale date and the calculated difference. The result should be sorted by sale date in ascending order.

```sql
CREATE TABLE Sales
(
    sale_date DATE,
    fruit VARCHAR(20),
    sold_num INT,
    PRIMARY KEY (sale_date, fruit)
);

INSERT INTO Sales
VALUES
('2020-05-01','apples',10),
('2020-05-01','oranges',8),
('2020-05-02','apples',15),
('2020-05-02','oranges',15),
('2020-05-03','apples',20),
('2020-05-03','oranges',0),
('2020-05-04','apples',15),
('2020-05-04','oranges',16);

select sale_date,
sum(case when fruit='apples' then sold_num else -1*sold_num end) difference
from Sales
group by sale_date
order by sale_date
```

Data Output Messages Notifications

Showing rows: 1 to 4 Page No:

	sale_date date	difference bigint
1	2020-05-01	2
2	2020-05-02	0
3	2020-05-03	20
4	2020-05-04	-1

Question 2

Experiment 06

Hard

(B) Problem Statement

A company stores hierarchical employee reporting information in a table. Each record represents a node in a hierarchy where:

- id represents the current node.
- p_id represents the parent node.

Classify every node into one of the following categories:

- **Root**: A node that does not have any parent.
- **Leaf**: A node that has a parent but does not have any children.
- **Inner**: A node that has both a parent and at least one child.

Display the node ID along with its category. Return the result sorted by node ID.

```sql
CREATE TABLE Tree
(
    id INT PRIMARY KEY,
    p_id INT
);

INSERT INTO Tree
VALUES
(1,NULL),
(2,1),
(3,1),
(4,2),
(5,2);


select t.id,  (case when t.p_id is NULL then 'Root'
    when t.id in (select distinct p_id from Tree) then 'Inner'
    else 'Leaf' end) as type
    from Tree t
```

Data Output Messages Notifications

SQL Showing rows: 1 to 5 Page No: 1

	id [PK] integer	type text
1	1	Root
2	2	Inner
3	3	Leaf
4	4	Leaf
5	5	Leaf